Project Report
on

ChurnLens: Analysing and Predicting Customer Attrition

Department of Statistics
Government Vidarbha Institute of Science and Humanities
(Autonomous),
Amravati

Submitted by
Sohel Firoj Shaikh
M.Sc. II (Sem - IV)
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Certificate

Department of Statistics
Government Vidarbha Institute of Science & Humanities
(Autonomous), Amravati
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This is certify that content in this report entitled **“ChurnLens: Analysing and Predicting Customer Attrition”** duly completed by **Mr. Sohel Firoj Shaikh** and submitted to the Department of Statistics, of Government Vidarbha Institute of Science and Humanities, Amravati (Autonomous) for the academic year 2024-25.

This report has not been submitted either partly or fully to any other university or institute for award of any degree or diploma.

Date : /04/2025
Place : Amravati

Dr. Neeta W. Andure
Head
Department of Statistics
G.V.I.S.H., Amravati

Declaration

I hereby declare that the work incorporated in this report entitled **"ChurnLens: Analysing and Predicting Customer Attrition"** in partial fulfilment of requirement for degree of Master of Science in Statistics is the outcome of original study undertaken by me and it has not been submitted earlier to any other university or institute for award of any degree or diploma.

Also, the data has not been derived from any thesis or publication of any university or scientific organization. The sources of material used and all assistance received during the investigation have been duly acknowledged.

Date : /04/2025
Place : Amravati

Sohel F. Shaikh
M.Sc. II
Department of Statistics
G.V.I.S.H., Amravati

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Sohel F. Shaikh
M.Sc. II
Department of Statistics
G.V.I.S.H., Amravati

Contents

Abstract

Customer churn is a significant challenge in the banking industry, where retaining existing customers is far more cost-effective than acquiring new ones. Understanding why customers leave a bank is vital for developing effective retention strategies. This project aims to explore and predict customer churn by applying a combination of statistical techniques, machine learning models, and data analysis methods.

We used a real-world dataset comprising demographic and behavioral details of banking customers. A comprehensive data pre-processing phase was carried out to ensure the reliability of the analysis. This included handling missing values, encoding categorical variables, detecting and retaining outliers using domain knowledge, applying Box-Cox transformation for skewed distributions, and dropping variables that lacked informative value.

Exploratory Data Analysis (EDA) helped uncover key trends and highlighted contrasts between customers who churned and those who did not. Correlation analysis provided insights into the strength and direction of relationships among numerical features. To address multicollinearity when independent variables are highly correlated we used the Variance Inflation Factor (VIF) to identify and mitigate redundant predictors, thereby improving the reliability of model interpretation.

In the multivariate analysis phase, clustering techniques such as the elbow method scores were used to group customers based on similarities in their features. This provided a deeper understanding of customer segments with higher churn risk. Additionally, survival analysis was applied to examine customer retention patterns over time, and Kaplan-Meier plots with log-rank tests were used to assess differences in survival curves across categorical groups like gender and geography.

For predictive modeling, we implemented both Logistic Regression (LR) and Extreme Gradient Boosting (XGBoost). These models were evaluated not only for performance metrics like recall and F1-score but also for their ability to identify influential features. Feature importance comparison revealed that while both models recognized key drivers of churn such as tenure, number of products, and credit score, XGBoost provided a more nuanced and ranked interpretation due to its ability to capture complex interactions between variables.

In conclusion, the study offers a data-driven perspective on the factors influencing bank customer churn and demonstrates how combining statistical, survival, and machine learning approaches can produce actionable insights. These findings can support banking institutions in crafting targeted, evidence-based strategies to improve customer retention and long-term engagement.

Keywords: Customer retention, churn, principal component, multivartiate analysis, clustering, logistic regression, feature importance.

Chapter 1
Introduction

In today's competitive world, customer retention has become more important than ever for businesses. Companies are no longer just trying to attract new customers they are also working hard to keep their current ones. This is especially true in industries like telecommunications, retail, and banking, where customers have many options to choose from. In such sectors, one of the biggest challenges faced by companies is customer churn.

Customer churn, also known as customer attrition, is the phenomenon where customers stop doing business with a company. In simple terms, it refers to the loss of customers over a period of time. For example, if a customer closes their bank account and moves to another bank, this is considered churn. Understanding why customers leave and how to reduce this number is extremely important for any business that wants to stay profitable and grow sustainably.

1.1 What is Customer Churn?

Customer churn is typically measured as the percentage of customers who stop using a company's product or service during a given time period. For instance, if a bank starts with 10,000 customers and loses 500 of them by the end of the month, its monthly churn rate is 5%. This number is not just a metric it represents lost revenue, missed opportunities, and often higher future costs, since gaining new customers usually requires more effort and money than retaining existing ones [11].

There are two main types of customer churn: voluntary churn and involuntary churn. Voluntary churn occurs when customers decide to leave on their own. This may be due to poor service, high fees, lack of features, or simply a better offer from a competitor. Involuntary churn happens when the company terminates the relationship, for example, due to non-payment or fraud. While involuntary churn is often out of a company's control, voluntary churn can be reduced with the right strategies—if companies can understand the reasons behind it.

1.2 Why is Customer Churn Important?

Customer churn has a direct impact on a company's financial performance. Losing customers means losing the revenue they bring. Research has shown that increasing customer retention by just 5% can increase profits by 25% to 95% [11]. This is because long-term customers not only buy more but are also more likely to recommend the company to others. They are less sensitive to price changes and are more forgiving of occasional service failures [7].

In banking, the importance of retaining customers is even higher. Banks offer many services like savings accounts, credit cards, loans, and investment options that build long-term relationships with customers. When a customer leaves a bank, the institution not only loses future business but also valuable customer data and the potential for cross-selling other services. Moreover, acquiring new customers in the financial sector is often more expensive due to strict regulations and intense competition [1].

High churn rates can also be a warning sign of bigger problems within the company, such as poor customer service, outdated technology, or ineffective communication. By studying churn, businesses can uncover these hidden issues and take steps to improve overall customer satisfaction.

1.3 What Causes Customer Churn?

Understanding what causes customers to leave is a crucial step in preventing churn. The reasons for churn can be complex and vary from one industry to another. However, some common causes include:

1. **Poor Customer Experience:** Customers today expect quick, easy, and personalized service. Long wait times, rude staff, or a confusing website can drive customers away.

2. **Better Offers from Competitors:** In a market where customers have many choices, even small differences in fees, interest rates, or rewards programs can influence their decision to switch.

3. **Lack of Engagement:** If a company does not actively stay in touch with its customers or fails to understand their changing needs, it risks losing their interest and loyalty.

4. **Product or Service Issues:** Bugs, delays, or unclear pricing can frustrate customers and lead them to try other providers.

5. **Life Events or External Factors:** Sometimes churn is caused by events outside the company's control, such as customers moving to a new city or facing a financial crisis.

For banks, additional reasons for churn may include security concerns, hidden fees, poor digital experience, or lack of trust. With increasing digitization and customer expectations, banks must continuously improve their services to stay competitive.

1.4 Why Predicting Churn Matters?

By predicting customer churn, businesses can take proactive steps to keep customers from leaving. Instead of waiting for customers to leave and then reacting, companies can identify those who are at high risk of leaving and try to convince them to stay through personalized offers, better service, or simple engagement.

Predictive models that analyze customer behavior, transaction history, complaints, and other patterns can help detect early warning signs. This is especially useful in sectors like banking, where customer relationships are long-term and complex [8].

For instance, if a bank notices that a customer who once used multiple services has stopped using them or is making fewer transactions, it could be a sign of potential churn. The bank could then contact the customer to offer help or suggest products tailored to their needs. Such timely action can go a long way in improving customer satisfaction and loyalty.

Predicting churn also helps in planning marketing efforts more efficiently. Instead of targeting all customers with the same message, companies can focus their resources on those who are most likely to leave, leading to better returns on investment.

1.4.1 The Growing Importance of Churn Analysis in Banking

The banking industry has witnessed major changes over the past decade. Digital transformation, online banking, fintech startups, and changing customer expectations have forced traditional banks to innovate constantly. With so many options available, customers are more willing than ever to switch banks if they feel unsatisfied [1].

In this environment, banks are investing heavily in understanding customer behavior. Churn analysis has become a key tool for improving customer experience, offering better financial products, and building long-term relationships.

Moreover, regulatory changes and competition from digital-only banks have made it necessary for traditional banks to prioritize customer retention. Churn prediction allows banks to identify patterns, trends, and customer segments that are more likely to leave, helping them design smarter strategies [8].

1.4.2 Social and Economic Impact of Churn

Beyond individual businesses, customer churn can also have a broader economic impact. High churn rates can lead to unstable revenues for companies, which can affect their ability to invest in employees, innovation, and growth. In financial services, sudden drops in customer numbers can even impact market confidence and stock prices.

From a social perspective, poor customer service and unfair practices leading to churn can erode trust in institutions especially banks, which play a key role in the economy. Therefore, reducing churn is not only a business goal but also contributes to building stronger, more responsible companies.

Chapter 2
Review of Literature

The study [4] presented a comprehensive approach for customer retention in the streaming service industry using a hybrid model that integrates big data, machine learning, and explainable AI. Their research highlights how combining deep learning with explainable AI improves the interpretability of churn prediction models and provides actionable insights for decision-makers. Though the study focuses on streaming services, the insights regarding model transparency and the use of complex data pipelines are highly relevant for the banking industry, where understanding why a customer might leave is just as important as predicting that they will.

The authors[12] proposed a predictive model for customer churn that uses historical customer data. Their model emphasized data preprocessing, feature selection, and the importance of using classification techniques to improve prediction accuracy. The study underscored the need for identifying potential churners early, allowing companies to intervene with retention strategies. This foundational work illustrates the evolution of churn modeling and its reliance on predictive analytics to guide customer relationship management strategies.

In the context of the banking industry, the authors[2] explored how machine learning can be leveraged to predict churn using structured data. The study used various classifiers and emphasized the impact of demographic and behavioral variables such as age, balance, and transaction history. It found that banking institutions can effectively use supervised learning models to identify at-risk customers, and by doing so, take measures to improve customer engagement. This study is particularly relevant to the present project, as it directly aligns with our goal of understanding churn dynamics in banks.

Another important aspect highlighted in churn studies is the treatment of data quality and the presence of outliers. The authors[9] stressed the importance of identifying and managing outliers in datasets. They argued that failing to handle outliers appropriately can distort statistical analyses and predictive model performance. Their findings are applicable to churn prediction as well, where outliers in features like customer balance or tenure could lead to misleading results. Acknowledging this, our study carefully analyzed outliers during the pre-processing phase and made informed decisions on whether to retain or exclude them based on domain understanding.

The authors[6] focused on customer churn in the banking sector using explainable machine learning models. Their work emphasized the use of techniques like SHAP (SHapley Additive exPlanations) to explain the contributions of each feature in the prediction process. Their results showed that features such as credit score, tenure, and balance significantly influence churn decisions. This approach not only improved prediction accuracy but also made the models more interpretable to stakeholders. This study supports the idea that integrating model explainability into churn analysis provides banks with more than just numbers—it offers strategic insights for retention

Chapter 3
Objectives

1. To examine the distribution of independent variables and assess their normality.
2. To uncover patterns and insights related to customer churn through exploratory data analysis (EDA).
3. To address multicollinearity among variables by applying Principal Component Analysis (PCA).
4. To develop and evaluate predictive models Logistic Regression and XGBoost for customer churn prediction, and to interpret feature importance from these models.
5. To identify groups of customers with similar characteristics using K-Means clustering.
6. To analyze the impact of different categories of categorical variables on customer churn using survival analysis techniques.

Chapter 4
Methodology

To achieve the above objectives, the following steps are followed in this study:

4.1 Data Collection

The bank customer churn dataset used in this study was downloaded from Kaggle which is one of the most popular data science competition platform and online community for data scientists and machine learning practitioners under Google LLC.

The dataset (available at `https://www.kaggle.com/datasets/rangalamahesh/bank-churn/data?select=train.csv`) was uploaded by the user RANGALA MAHESH. It contains two csv files for training and testing purpose. In this study we particularly used the training dataset as it contains the infromation about churned customers.

4.2 Data Description

The dataset consists of 165034 rows and 14 columns. Each row represents a unique customer and each column represents some information about the corresponding customer.

The columns (features) in the dataset are as follows:

Column Name	**Description**
id	Row number
CustomerId	Unique ID assigned to each customer
Surname	Surname of the customer
CreditScore	Customer's credit score (range: 350-850)
Geography	Country of the customer (France, Spain, Germany)
Gender	Male or Female
Age	Customer's age (in years)
Tenure	Number of years the customer has been with the bank
Balance	Bank balance of the customer
NumOfProducts	Number of products (services) the customer has purchased
HasCrCard	Whether the customer owns a credit card (0 = No, 1 = Yes)
IsActiveMember	Whether the customer is an active member (0 = No, 1 = Yes)
EstimatedSalary	Customer's estimated salary
Exited	Whether the customer left the bank (0 = No, 1 = Yes)

Table 4.1: Description of Customer Dataset Columns

4.3 Data Pre-processing

Before analyzing any dataset, we need to clean and prepare it properly. This process is called data pre-processing. It ensures that the data is free from errors, missing values, and inconsistencies. We pre-processed the data in python using following steps:

4.3.1 Importing required libraries

The first step in data preprocessing is to import the necessary libraries. These libraries help us in loading, cleaning, and analyzing the dataset. The libraries used are:

```
import pandas as pd # for handling and processing the dataset
# For visualization
import seaborn as sns
import matplotlib.pyplot as plt
```

These libraries make it easier to work with large amounts of data efficiently.

4.3.2 Loading the dataset

After importing the libraries, the dataset is loaded using the pandas library. The following command is used to read the dataset:

```
df = pd.read_csv("train.csv")
```

4.3.3 Removing unnecessary columns

The columns “id” and “Surname” are not useful for the analysis. So we removed these columns using:

```
df = df.drop([‘id’, ‘Surname’, ], axis = 1)
```

4.3.4 Checking for missing values

Missing values can cause errors in analysis. We checked for missing values using:

```
df.isnull().sum()
```

Which gives the output:

```
Output:
CustomerId        0
CreditScore       0
Geography         0
Gender            0
Age               0
Tenure            0
Balance           0
NumOfProducts     0
HasCrCard         0
IsActiveMember    0
EstimatedSalary   0
Exited            0
dtype: int64
```

This showed that there were no missing values in the dataset.

4.3.5 Checking and removimg duplicate entries

We checked for duplicate rows (same data repeated):

```
df.duplicated().sum()
```

Which give the output as:

```
30
```

Therefore, we removed those duplicate entries using:

```
df = df.drop_duplicates()
```

4.3.6 Transforming the Tenure column

The "Tenure" column represents a customer's relationship with the bank in years. To make it more detailed, we convert it into days:

```
df['Tenure'] = df['Tenure']*365
```

4.3.7 Checking for validity of the data

To ensure that the dataset is correct and does not contain mistakes, we checked:

1. Unique values in Geography

   ```
   df['Geography'].unique()
   ```

   Which gave the output:

   ```
   array(['France', 'Spain', 'Germany'], dtype=object)
   ```

   This indicates that the dataset is about bank customers from the countries: France, Germany and Spain.

2. Unique values in Gender

   ```
   df['Gender'].unique()
   ```

   Which gave the output:

   ```
   array(['Male', 'Female'], dtype=object)
   ```

   This shows that the dataset contains the information about Female and Male customers.

3. Unique values in NumOfProducts

   ```
   df['NumOfProducts'].unique()
   ```

   Which gave the output:

   ```
   array([2, 1, 3, 4], dtype=int64)
   ```

   This indicates that the bank is offering 4 products (services).

4. Unique values in HasCrCard, IsActiveMember, and Exited

   ```
   df['HasCrCard'].unique()
   df['IsActiveMember'].unique()
   df['Exited'].unique()
   ```

   Which gave the output:

   ```
   array([1., 0.])
   array([0., 1.])
   array([0, 1], dtype=int64)
   ```

These are binary variables using 1 for Yes and 0 for No.

5. CreditScore Range

```
a = min(df['CreditScore'])
b = max(df['CreditScore'])
print("Minimum credit score = ", a)
print("Maximum credit score = ", b)
```

Which gave the output:

```
Minimum credit score =  350
Maximum credit score =  850
```

The valid range for credit scores is 300 - 850. The credit scores in the dataset are in the range 350 - 850. Which is acceptable.

6. Age Range

```
l = min(df['Age'])
m = max(df['Age'])
print("Minimum age = ", l)
print("Maximum age = ", m)
```

Which gave the output:

```
Minimum age =  18.0
Maximum age =  92.0
```

This shows that customers in the dataset are in the range 18 - 92 years.

4.3.8 Checking data types :

Using correct data types for the features is very important in the analysis. It is crucial for machine learning model selection also. We checked each feature's data type using:

```
df.info()
```

Which gave the output as:

```
<class 'pandas.core.frame.DataFrame'>
Index: 165002 entries, 0 to 165033
Data columns (total 12 columns):
 #   Column           Non-Null Count   Dtype
---  ------           --------------   -----
 0   CustomerId       165002 non-null  int64
 1   CreditScore      165002 non-null  int64
 2   Geography        165002 non-null  object
 3   Gender           165002 non-null  object
 4   Age              165002 non-null  float64
 5   Tenure           165002 non-null  int64
 6   Balance          165002 non-null  float64
 7   NumOfProducts    165002 non-null  int64
 8   HasCrCard        165002 non-null  float64
 9   IsActiveMember   165002 non-null  float64
 10  EstimatedSalary  165002 non-null  float64
 11  Exited           165002 non-null  int64
dtypes: float64(5), int64(5), object(2)
memory usage: 16.4+ MB
```

Here, the columns 'Age', 'HasCrCard' and 'EstimatedSalary' are saved as float points (reals) but which are supposed to be integers. So we convert then using:

```
numeric_cols = ['Age', 'HasCrCard', 'IsActiveMember'] # List of columns to
    change
# Changing datatype
df[numeric_cols] = df[numeric_cols].astype(int)
```

4.3.9 Encoding categorical variables :

Since, machine learning models do not understand text-based categories, we convert them into numbers as,

- Geography : (i) France → 1 (ii) Germany → 2 (iii) Spain → 3
- Gender : (i) Female → 1 (ii) Male → 2

Using:

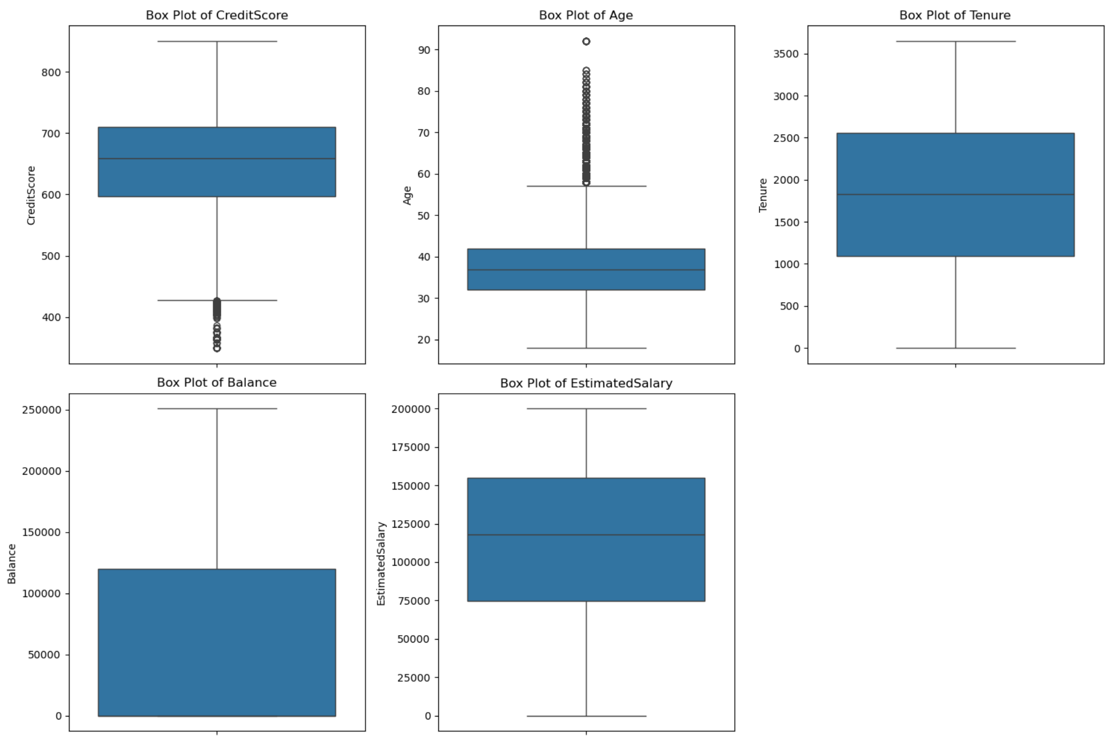
```
df['Geography'] = df['Geography'].replace({'France':1, 'Germany':2, 'Spain
    ':3})
df['Gender'] = df['Gender'].replace({'Female':1, 'Male':2})
```

4.3.10 Indentifying outliers :

Outliers are extreme values that can affect analysis[3]. We used box plots to detect outliers in numerical columns:

```
# List of columsn that might contain outliers
columns = ['CreditScore', 'Age', 'Tenure', 'Balance', 'EstimatedSalary']

plt.figure(figsize=(15, 10))  # Adjust figure size for all subplots
for i, var in enumerate(columns, 1):
    plt.subplot(2, 3, i)  # 2 rows, 3 columns, ith subplot
    sns.boxplot(y=df[var])  # Use 'y' to make the box plots vertical
    plt.ylabel(var)
    plt.title(f"Box Plot of {var}")
plt.tight_layout()
plt.show()
```

Which gives the plot:



The plot indicates that outliers are present in the columns: "Age" and "CreditScore".

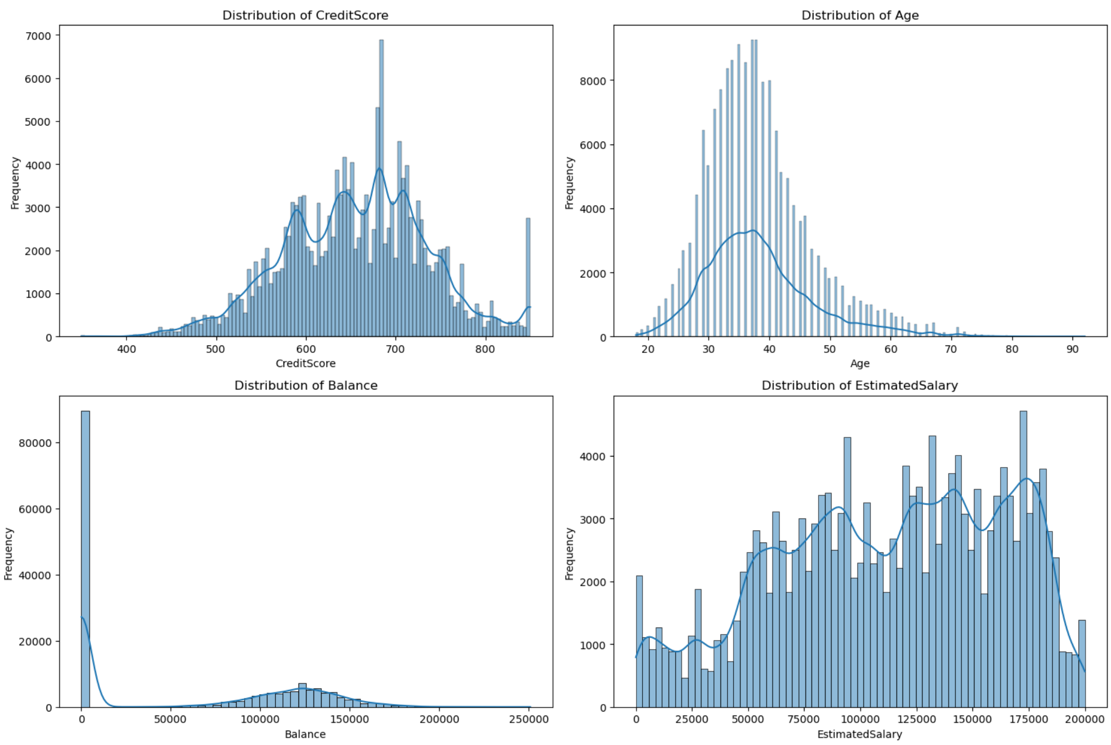
The box plot for Age indicates some outliers on the higher side (above 60). Since the dataset is about bank customers from France, Germany, and Spain, where life expectancy is over 80 years, these values are reasonable. Many banks provide services to elderly customers, and their presence in the data is not unusual. So these outliers are considered as valid, and decided to keep them for the analysis.

The box plot for Credit Score shows very few outliers on the lower side (below 420). Since the credit score typically ranges from 300 to 850, very low scores are rare but possible. Most of the values fall between 550 and 800, which means that the majority of customers have fair to excellent credit scores. The presence of a few low credit scores may represent customers who have a poor repayment history or limited credit history. Since these low scores are expected in real-world banking data, we decided to keep these outliers.

4.3.11 Checking for normality :

Normality means that the data follows a bell-shaped curve (normal distribution), which is important for many statistical tests and machine learning models. We plot bar graphs with KDE (Kernel Density Estimate) to visualize the distribution of numeric features.

```
# Checjing for normality
numerical_vars = ['CreditScore', 'Age', 'Balance', 'EstimatedSalary']

# Using Seaborn (combined plot)
plt.figure(figsize=(15, 10))  # Adjust figure size for all subplots
for i, var in enumerate(numerical_vars, 1): #Enumerate to get index for
    subplots
    plt.subplot(2, 2, i) # 2 rows, 2 columns, ith subplot
    sns.histplot(df[var], kde=True) #kde adds a Kernel Density Estimate
    plt.xlabel(var)
    plt.ylabel("Frequency")
    plt.title(f"Distribution of {var}")
plt.tight_layout() # Adjust layout to prevent labels from overlapping
plt.show()
```



From the plots:

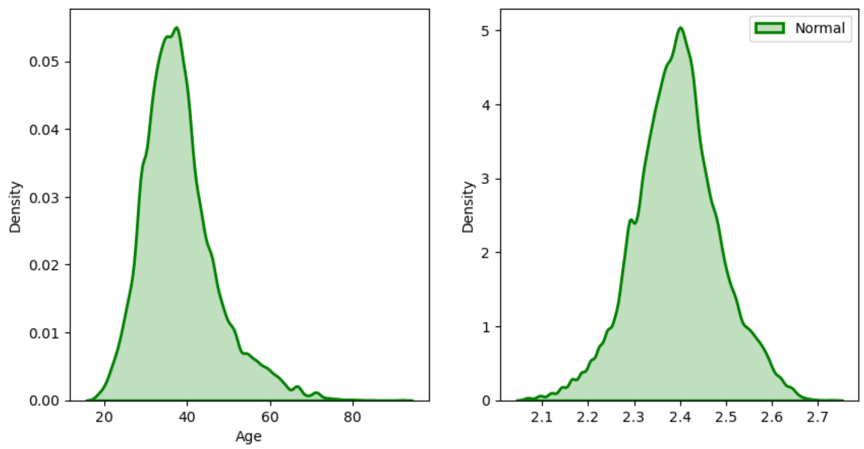
- The "CreditScore" column is almost normally distributed, so no transformation is required.
- The column "Age" is right-skewed implying that there are more younger individuals and fewer older individuals. To reduce skewness and normalize the column 'Age' we use Box-Cox transformation as :

$$T(y) = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \ln(y), & \lambda = 0 \end{cases}$$

```
data = df['Age']

# transform training data & save lambda value
fitted_data, fitted_lambda = stats.boxcox(data)

# creating axes to draw plots
fig, ax = plt.subplots(1, 2)

# plotting the original data(non-normal) and
# fitted data (normal)
sns.distplot(data, hist = False, kde = True,
      kde_kws = {'shade': True, 'linewidth': 2},
      label = "Non-Normal", color ="green", ax = ax[0])

sns.distplot(fitted_data, hist = False, kde = True,
      kde_kws = {'shade': True, 'linewidth': 2},
      label = "Normal", color ="green", ax = ax[1])

# adding legends to the subplots
plt.legend(loc = "upper right")

# rescaling the subplots
fig.set_figheight(5)
fig.set_figwidth(10)

print(f"Lambda value used for Transformation: {fitted_lambda}")
```

Which give the output as:

```
Lambda value used for Transformation: -0.24589289915500895
```



Hence, we get $\lambda = -0.24589$. We use this value to transform 'Age' as:

```
df['transf_Age'] = (df['Age']**-0.24589 - 1)/(-0.24589)
```

This will create a new transformed column named 'transf_Age' in the dataframe using Box-Cox transformation.

- The Balance column should be dropped because it has over 80,000 zero values, meaning most customers have no balance. When a column has too many zeroes, it does not add useful information to the model and may reduce its accuracy and increases training time significantly[10]. In machine learning, features with very little variation do not help in making predictions and can even make the model more complex without improving results. Transforming this column is also difficult because methods like log transformation cannot be applied to zero values properly. We drop the columns using:

```
df = df.drop("Balance", axis=1)
```

- The distribution of column 'EstimatedSalary' is uniformly spread, which means salaries are evenly distributed. It does not show significant skewness, so transformation is not necessary.

In this section, we cleaned and prepared the data for further analysis. We handled missing values and removed duplicate entries. Categorical data was converted into numerical form so that it could be used in models. We identified outliers in the 'Age' and 'CreditScore' columns but decided to keep them because they represent real customer behaviors rather than errors. To improve the distribution of the 'Age' column, we applied the Box-Cox transformation. We also removed the 'Balance' column because it had too many zero values, making it less useful for analysis. With the dataset now ready, we can move on to Exploratory Data Analysis (EDA) to find patterns and relationships in the data.

Chapter 5
Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a crucial step in understanding the dataset before applying any statistical technique or machine learning model. It helps in identifying patterns, relationships, and anomalies in the data. In this section, we perform a detailed examination of the dataset through various statistical and visualization techniques.

5.1 Univariate Analysis:

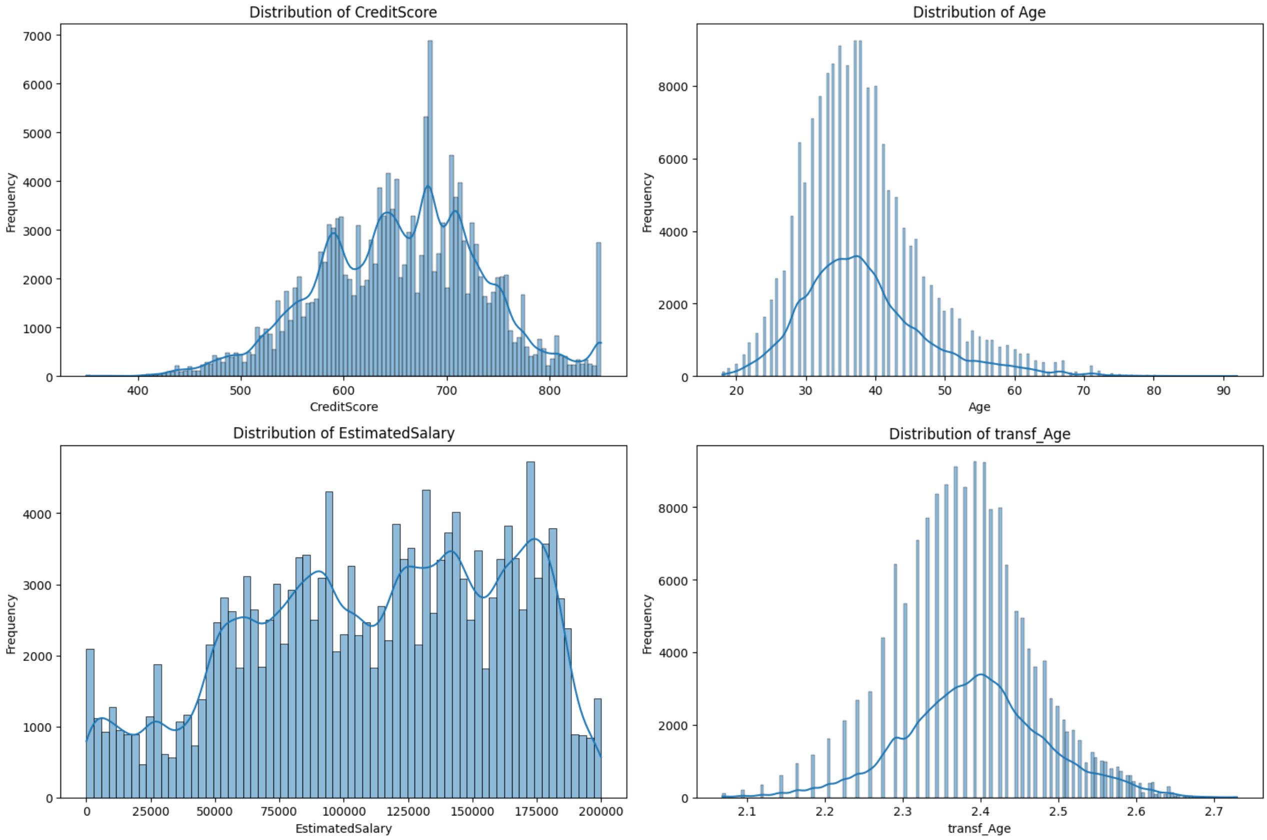
Univariate Analysis is an essential technique in Exploratory Data Analysis (EDA). It allows us to understand individual variables in a dataset. It helps uncover patterns, anomalies, and insights that can drive decision-making.

5.1.1 Numerical Variables:

In the dataset, the features (variables) 'CreditScore', 'Age', 'Tenure', 'EstimatedSalary', 'transf_Age' are numerical features. We analyze them by finding their distribution using KDE (Kernel Density Estimate) enabled bar charts and find the descriptive statistic as:

```
# Bar plot
num_var = ['CreditScore', 'Age', 'EstimatedSalary', 'transf_Age']
plt.figure(figsize=(15, 10))  # Adjust figure size for all subplots
for index, var in enumerate(num_var):
    plt.subplot(2, 3, index + 1)
    sns.histplot(df[var], kde=True) #kde adds a Kernel Density Estimate
    plt.xlabel(var)
    plt.ylabel("Frequency")
    plt.title(f"Distribution of {var}")
plt.tight_layout()
plt.show()

df_num = df[num_var]
df_num.describe()
```

```
Output:
       CreditScore Age      EstimatedSalary  transf_Age
count 165002       165002    165002           165002
mean  656.457661   38.125405 112575.712332    2.392562
std   80.101454    8.866964  50293.089358     0.091878
min   350.000000   18.000000 11.580000        2.068838
25%   597.000000   32.000000 74637.595000     2.332425
50%   659.000000   37.000000 117948.000000    2.393250
75%   710.000000   42.000000 155152.467500    2.444607
max   850.000000   92.000000 199992.480000    2.729083
```



Conclusion: The univariate analysis of numerical variables provides important insights into the dataset. The CreditScore ranges from 350 to 850, with an average of 656.46, showing a nearly normal distribution with some peaks, which suggest possible clusters of customers with similar credit behavior. The Age variable has a mean of 38.12 years and a median of 37 years, indicating a slightly right-skewed distribution. Most customers fall between 32 and 42 years, and to improve normality, a Box-Cox transformation was applied, resulting in the transf_Age variable, which now appears more symmetric.

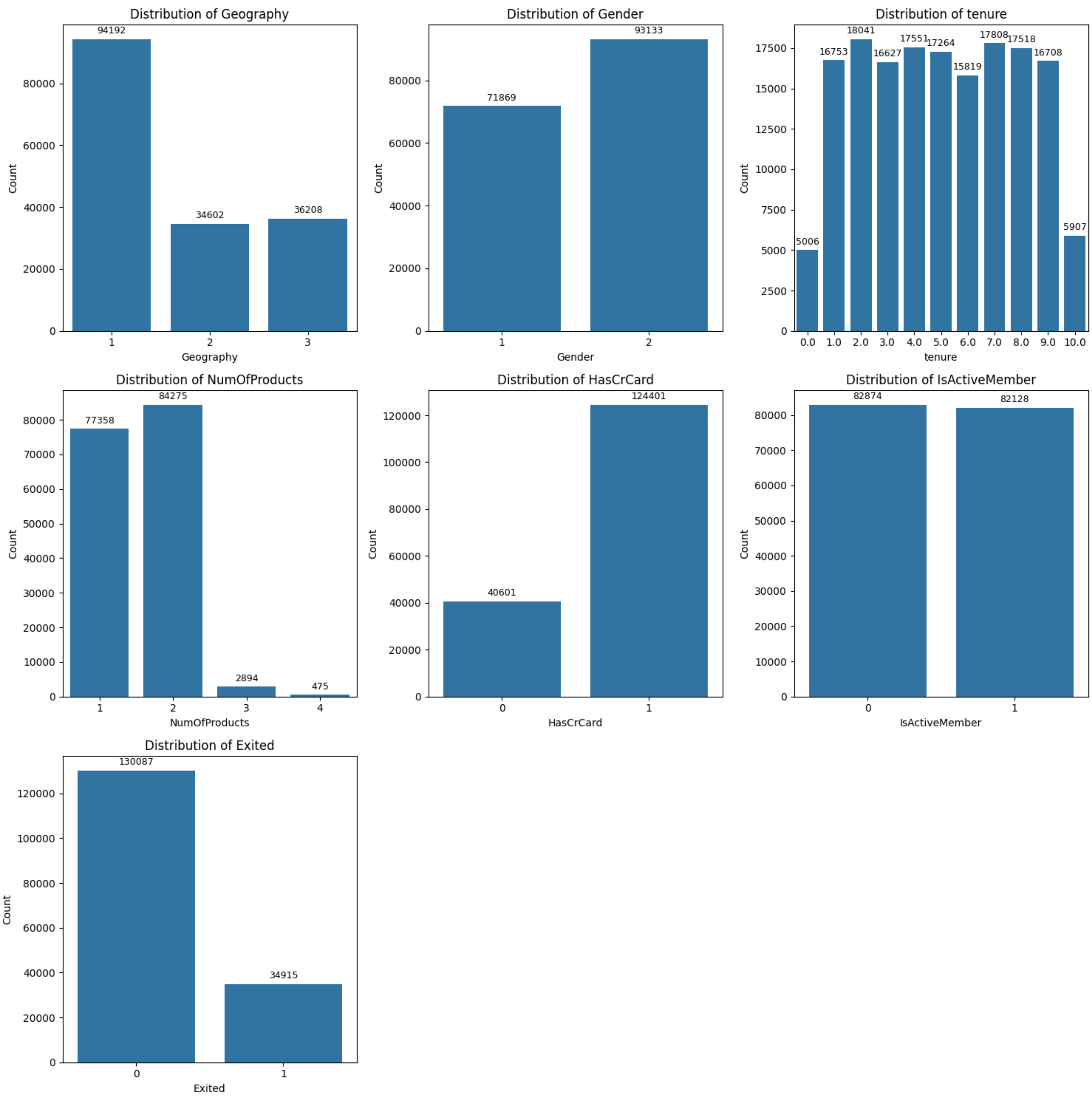
The EstimatedSalary variable varies widely, from €11.58 to €199,992.48, with an average of €112,575.71. The distribution appears uniform, meaning salaries are spread evenly across different income levels without a strong pattern.

5.1.2 Categorical Variables:

The categorical variables in the dataset include Geography, Gender, Tenure, Number of Products, Credit Card Ownership (HasCrCard), Active Membership Status (IsActiveMember), and Churn Status (Exited). The distribution of these categorical variables is visualized using bar plots as:

```
cat_cols = [‘Geography’, ‘Gender’, ‘tenure’, ‘NumOfProducts’, ‘HasCrCard’,
    ‘IsActiveMember’, ‘Exited’] # Tenure is re-transformed to years &
   denoted by tenure

plt.figure(figsize=(15, 15))
for i, col in enumerate(cat_cols):
    plt.subplot(3, 3, i + 1)
    ax = sns.countplot(data=df, x=col)
    plt.title(f"Distribution of {col}")
    plt.xlabel(col)
    plt.ylabel("Count")

```

```
    for p in ax.patches:
        height = p.get_height()
        if pd.notna(height):
            ax.annotate(f'{int(height)}', (p.get_x() + p.get_width() / 2., height),
                        ha='center', va='center', xytext=(0, 8), textcoords='offset points',
                        fontsize=9)
plt.tight_layout()
plt.show()
```



The categorical variables in the dataset show distinct distributions. Most customers belong to category 1 (France), followed by category 3 (Spain) and category 2 (Germany). In terms of gender, there are more male customers (category 2) compared to female customers (category 1). The tenure variable is evenly distributed across different years, but a slight variation is observed in some values. The number of products used by customers is mainly

concentrated in categories 1 and 2, while very few customers fall into categories 3 and 4. A majority of customers possess a credit card (category 1), while a smaller portion does not (category 0). Similarly, the number of active and inactive members is nearly equal. Lastly, the churn variable shows that most customers did not leave the bank (category 0), whereas a smaller percentage of customers exited (category 1).
To determine whether these categorical variables significantly impact customer churn, we conducted a Chi-Square test of independence. The hypothesis to be tested is:

H_0: There is no significant relationship between the categorical variable and churn.

H_1: There is a significant relationship between the categorical variable and churn.

```
# Checking whether the categorical variable is significantly related to churn
for col in cat_cols:
    contingency_table = pd.crosstab(df[col], df["Exited"])
    chi2, p, dof, expected = chi2_contingency(contingency_table)
    print(f"Chi-Square Test for {col}: p-value = {p}")
```

```
Output:
Chi-Square Test for Geography: p-value = 0.0
Chi-Square Test for Gender: p-value = 0.0
Chi-Square Test for tenure: p-value = 3.129508151879632e-51
Chi-Square Test for NumOfProducts: p-value = 0.0
Chi-Square Test for HasCrCard: p-value = 2.8978197417288648e-19
Chi-Square Test for IsActiveMember: p-value = 0.0
Chi-Square Test for Exited: p-value = 0.0
```

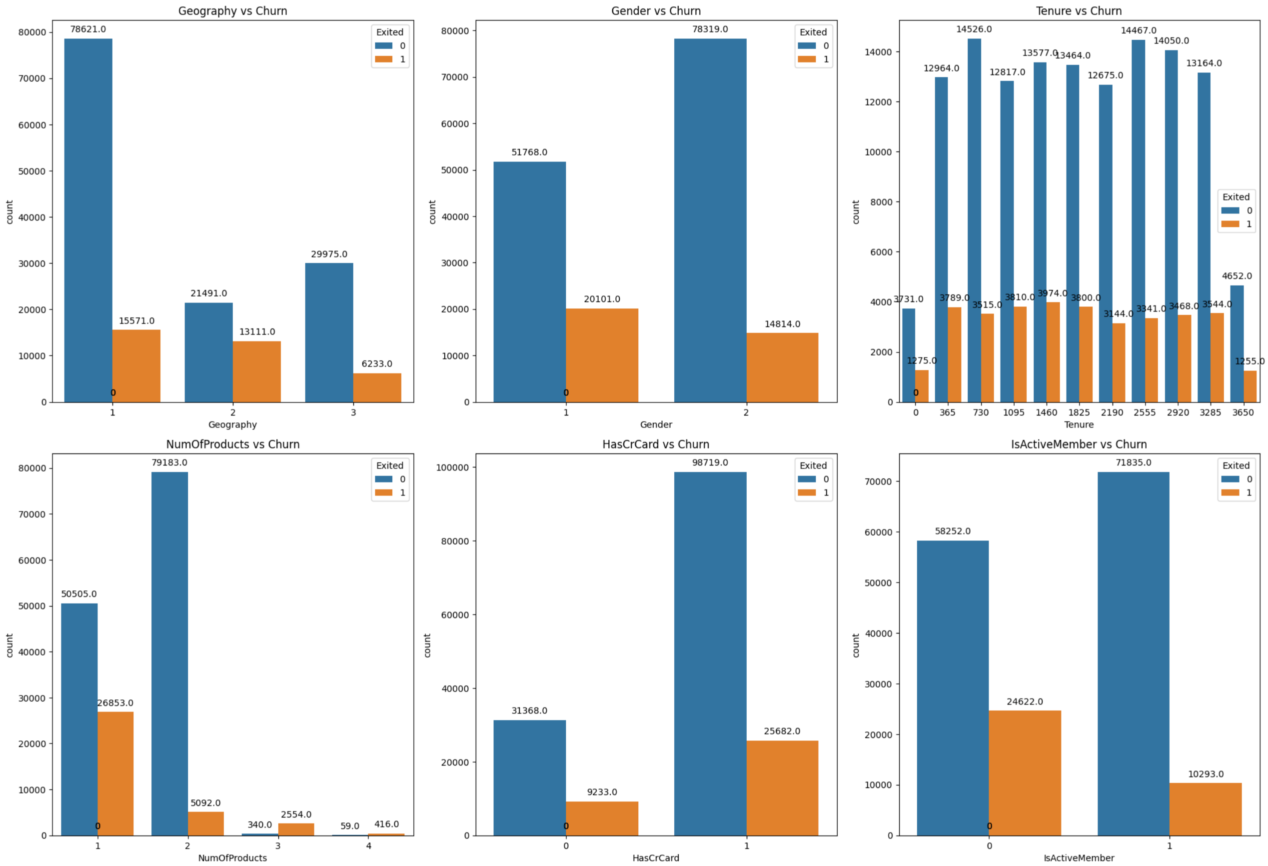
Since the p-values for all categorical variables are less than the significance threshold of 0.05, we reject H_0 at 5% level of significance and conclude that all these variables have a statistically significant relationship with customer churn. This means that customer churn is influenced by factors such as location, gender, tenure, product ownership, credit card status, and active membership status.

5.2 Bivariate Analysis:

To understand the factors influencing customer churn, we analyzed the relationship between different categorical and numerical variables with churn. This analysis helps in identifying patterns that may contribute to customer attrition. Categorical variables such as Geography, Gender, Tenure, Number of Products, Credit Card Ownership, and Activity Status were compared against churn to observe significant trends.

```
# List of categorical columns you want to plot
categorical_columns = ['Geography', 'Gender', 'Tenure',     'NumOfProducts', 'HasCrCard', 'IsActiveMember']

plt.figure(figsize=(20, 20))
for i, column in enumerate(categorical_columns):
    plt.subplot(3, 3, i + 1)
    ax = sns.countplot(x=column, hue='Exited', data=df)
    plt.title(f'{column} vs Churn')

    for p in ax.patches:
        ax.annotate(f'{p.get_height()}', (p.get_x() + p.get_width() / 2., p.get_height()),
                    ha='center', va='center', xytext=(0, 10), textcoords='offset points',
```

```
                          fontsize=8)

plt.tight_layout()
plt.show()
```



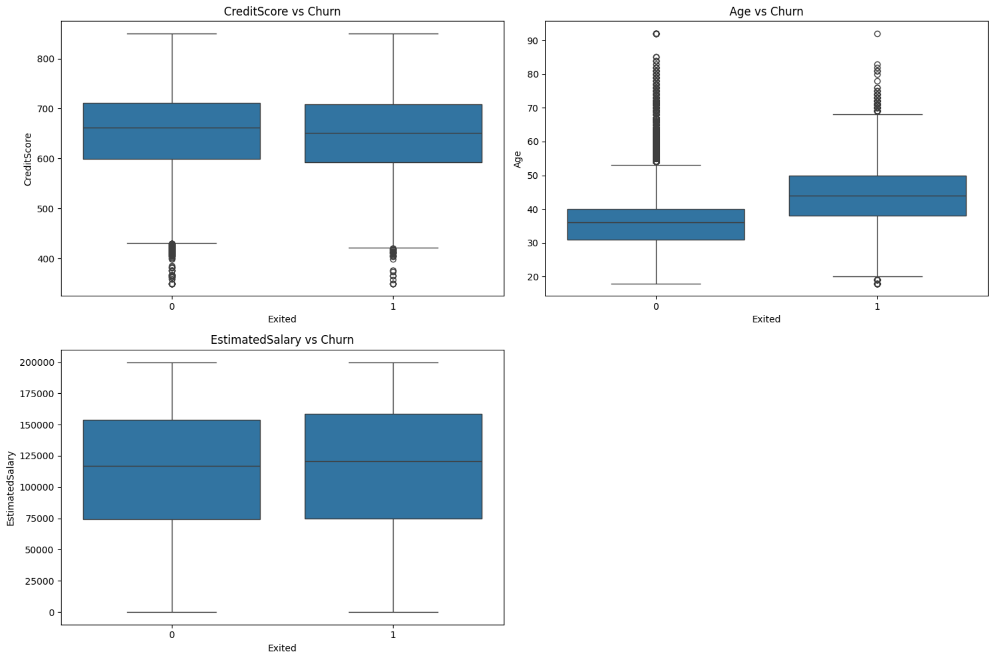
Key findings:

- Geography: Customers from Germany have a higher churn rate compared to France and Spain.
- Gender: Female customers have higher churn rate than male customers.
- Tenure: Churn rates fluctuate across different tenure levels, indicating that tenure might influence customer retention.
- Number of Products: Customers with only one product are more likely to leave, while those with 2 and 3 products have a lower churn rate. Customers with four products have a significantly higher likelihood of leaving the bank.
- Credit Card Ownership: Customers with credit cards are less likely to churn than those who do not have credit card.
- Activity Status: Inactive members have a much higher churn rate, confirming that customer engagement is a crucial factor.

These findings highlight the importance of demographic and behavioral factors in predicting customer churn.

Similarly, numerical variables like Credit Score, Age, and Estimated Salary were examined to assess their impact on customer retention.

```python
columns_to_plot = [‘CreditScore’, ‘Age’, ‘EstimatedSalary’]

plt.figure(figsize=(15, 10))
for i, column in enumerate(columns_to_plot):
    plt.subplot(2, 2, i + 1)
    sns.boxplot(x=‘Exited’, y=df[column], data=df)
    plt.title(f‘{column} vs Churn’)
    plt.ylabel(column)

plt.tight_layout()
plt.show()
```

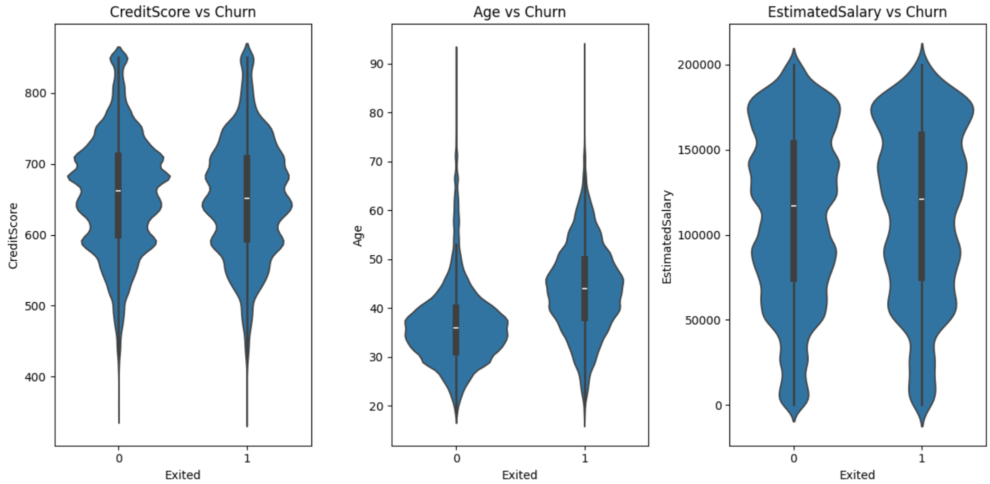


From the above plots, we can conclude that credit scores appear similar between churned and retained customers, indicating a weak relationship. However, older customers show a higher tendency to churn, suggesting age is an important factor. Estimated salary does not seem to have a strong influence on customer churn.

To understand how numerical variables vary between customers who stayed and those who left, we used violin plots. These plots show the distribution of numerical features across churn categories. The spread and density of the data in these plots help us see if churned customers differ significantly from non-churned customers.

```python
# Distribution using violin plot with respect to churn
num_cols = [‘CreditScore’, ‘Age’, ‘EstimatedSalary’]
plt.figure(figsize=(12, 6))
for i, column in enumerate(num_cols):
    plt.subplot(1, 3, i + 1)
    sns.violinplot(x=‘Exited’, y=df[column], data=df)
    plt.title(f‘{column} vs Churn’)
    plt.ylabel(column)

plt.tight_layout()
plt.show()
```

This gives the output plot as:



Key observations:

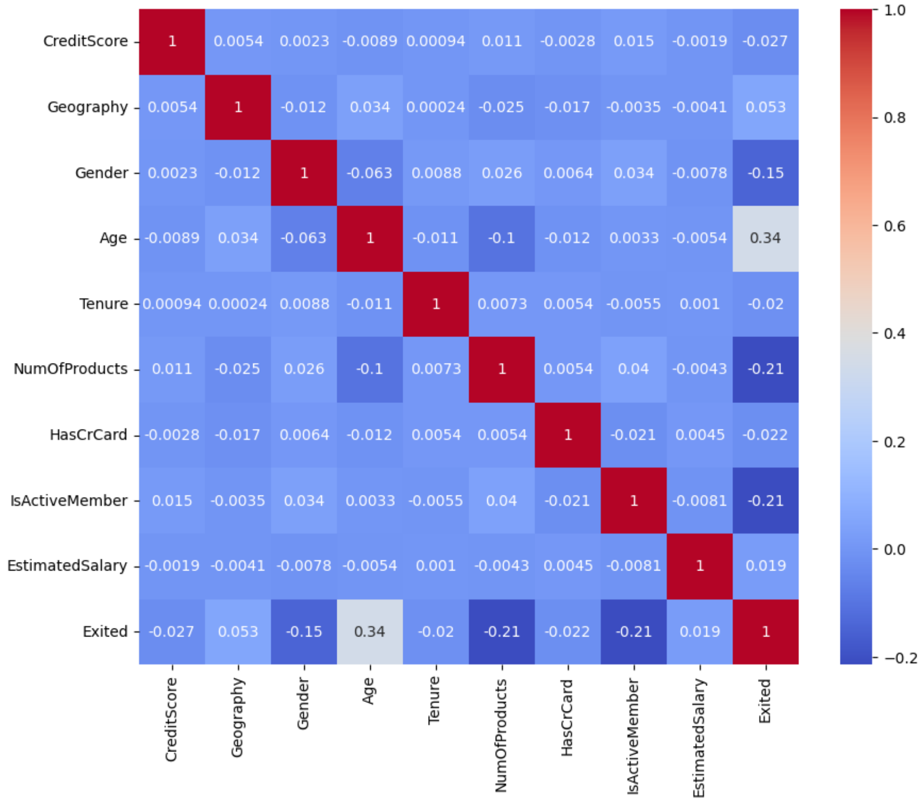
- Age: Customers who churned tend to have a higher median age compared to those who stayed. This suggests older customers might be more likely to leave.
- Credit Score: There is no strong difference in credit score distribution between churned and non-churned customers. This indicates that credit score alone may not be a strong predictor of churn.
- Estimated Salary: The distribution of salaries is similar across both churn categories, suggesting that salary has little impact on churn behavior.

5.3 Correlation Analysis:

In this section, we examine how different variables in the dataset relate to one another. Correlation helps us understand whether changes in one feature are associated with changes in another. By analyzing these relationships, we can identify which factors might have the most influence on customer churn.

We use a correlation heatmap to visualize the strength and direction of relationships between variables.

```
# Correlation analysis
cols = [‘CustomerId’, ‘transf_Age’]
df1 = df.drop(cols, axis=1)

df1 = pd.DataFrame(df1)
corr = df1.corr()
plt.figure(figsize=(10, 8))
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.show()
```



Key Observations

1. **Age and Churn (0.34):** Age has the strongest positive correlation with churn among all features, meaning older customers are more likely to leave. This suggests that senior customers might have different expectations or banking needs that influence their decision to exit.

2. **Number of Products and Churn (-0.21):** There is a moderate negative correlation, indicating that customers with more products are less likely to churn. This suggests that cross-selling additional banking products (such as loans, credit cards, or savings accounts) might help retain customers.

3. **IsActiveMember and Churn (-0.21):** Active customers are significantly less likely to churn. This highlights the importance of customer engagement—regularly interacting with the bank (e.g., transactions, inquiries, or account usage) reduces the chances of leaving.

4. **Gender and Churn (-0.15):** The negative correlation suggests that female customers are more likely to churn compared to male customers. Although the effect is not very strong, it could indicate different banking preferences or levels of satisfaction among genders.

5. **Geography and Churn (0.05):** Geography shows a slight correlation with churn, meaning that customers from Germany and Spain may have higher churn rates. This might be due to regional banking policies, service quality, or customer preferences.

6. **Credit Score and Churn (-0.03):** The weak negative correlation indicates that credit score has little impact on churn. A lower credit score does not necessarily mean a higher churn probability, implying that customers do not leave just because of their creditworthiness.

7. **Estimated Salary and Churn (0.02):** Salary has almost no correlation with churn, meaning income level does not play a major role in customer retention. Customers with both high and low salaries have a similar likelihood of staying or leaving.

8. **Other Variables:** Features like tenure, having a credit card, and other banking details have very weak correlations with churn, suggesting they are not strong predictors.

The Exploratory Data Analysis (EDA) provided valuable insights into the factors influencing customer churn. The univariate analysis helped us understand the distribution of individual features, while bivariate analysis revealed relationships between numerical and categorical variables with churn. We found that older customers, those with fewer banking products, and inactive members are more likely to churn, highlighting the importance of customer engagement and cross-selling. The correlation analysis further confirmed that age, number of products, and activity status are the strongest predictors of churn, while credit score and estimated salary have little influence.

Chapter 6
Multivariate Analysis

In this section, we analyze how multiple variables interact with each other to influence customer churn. Unlike univariate and bivariate analysis, which focus on individual or pair-wise relationships, multivariate analysis helps us understand complex dependencies among features. We will first check for multicollinearity to ensure that variables are not highly correlated, which can affect analysis. If necessary, we will apply Principal Component Analysis (PCA) to reduce dimensionality. We will then use Logistic Regression and XGBoost to determine feature importance, highlighting the key factors that drive churn. Additionally, we will apply K-Means Clustering to identify customer segments with similar characteristics. This comprehensive approach will give us deeper insights into customer behavior.

6.1 Multicollinearity:

Multicollinearity occurs when independent variables in a dataset are highly correlated with each other, making it difficult to determine their individual effect on the dependent variable. To detect multicollinearity, we calculate the Variance Inflation Factor (VIF) for each feature. A high VIF value (typically above 10) indicates strong multicollinearity.

```
from statsmodels.stats.outliers_influence import variance_inflation_factor

cols = ["CustomerId", "Age", "Exited"]
df1 = df.drop(cols, axis = 1)

# Calculate VIF for each selected column
vif_data = pd.DataFrame()
vif_data["Feature"] = df1.columns
vif_data["VIF"] = [variance_inflation_factor(df1.values, i) for i in range
    (df1.shape[1])]

print(vif_data)
```

```
Output:
            Feature        VIF
0       CreditScore  62.358100
1         Geography   5.074670
2            Gender  10.731148
3            Tenure   4.182410
4     NumOfProducts   8.873088
5         HasCrCard   4.047895
6    IsActiveMember   1.996047
7   EstimatedSalary   5.969017
8        transf_Age  85.520837
```

The output revealed that Transformed Age (VIF = 85.52) and Credit Score (VIF = 62.36) have extremely high multicollinearity, indicating that these variables share a significant amount of information with other features. Additionally, Gender (VIF = 10.73) also shows moderate multicollinearity, suggesting some redundancy in the dataset. NumOfProducts (VIF = 8.87) is slightly below the threshold but still noteworthy. Other variables have VIF values within an acceptable range.

To address multicollinearity and improve the stability of the analysis, we will apply Principal Component Analysis (PCA).

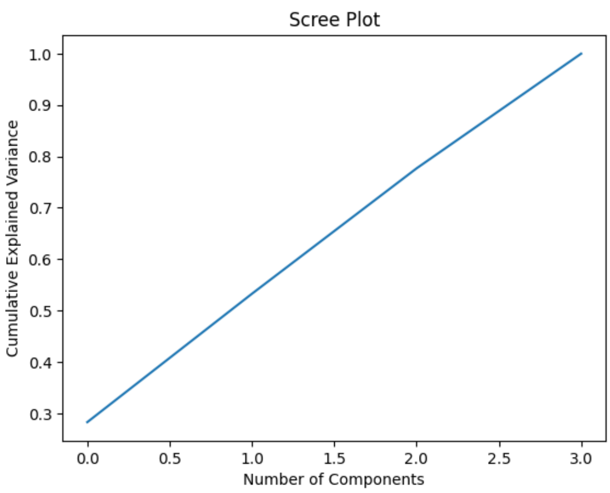
6.2 Principal Component Analysis (PCA):

Principal Component Analysis (PCA) is a dimensionality reduction technique used to address multicollinearity and improve the efficiency of models. It works by transforming correlated variables into a new set of uncorrelated variables, called principal components, which capture the most important patterns in the data.

To find principal components, we have used code:

```
multcol_cols = ["CreditScore", "Gender", "transf_Age", "NumOfProducts"] # columns with multicollinearity

# Standardizing the data
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
scaled_data = scaler.fit_transform(df1[multcol_cols])   #df1 is the dataframe with columns CustomerId, Age and Exited

from sklearn.decomposition import PCA
# Initialize PCA (set n_components=None initially)
pca = PCA(n_components=None)
pca.fit(scaled_data)
# Transform the data into principal components
pca_transformed = pca.transform(scaled_data)

# Analyzing PCA results by plotting the explained variance ratio
plt.plot(np.cumsum(pca.explained_variance_ratio_))
plt.xlabel(‘Number of Components’)
plt.ylabel(‘Cumulative Explained Variance’)
plt.title(‘Scree Plot’)
plt.show()
```

This gave the Scree plot[1]:



This shows that component-1 explains 33.33% variance, component-2 explains 66.67%

[1]Scree plots are graphical tools used in multivariate analysis, particularly in Principal Component Analysis (PCA), to visualize the amount of variance explained by each principal component. They help to determine the optimal number of components to retain by displaying the eigenvalues in descending order.

variance and component-3 explains 100% variance. Since maximum variance is explained by component-3, we use all three components.

Then we calculate load of features on the principle components (PC) using:

```
# Calculating three principle componenets
pca = PCA(n_components=3)
principal_components = pca.fit_transform(scaled_data)

# Interpretation
loadings = pd.DataFrame(
    pca.components_.T,
    columns=[f'PC{i+1}' for i in range(pca.n_components_)],
    index=df[multcol_cols].columns
)

print(loadings)
```

```
Output:
                    PC1        PC2        PC3
CreditScore   -0.106784   0.973392   0.202549
Gender        -0.426922  -0.218840   0.838073
transf_Age     0.665027   0.053653   0.125177
NumOfProducts -0.603386   0.041706  -0.490854
```

Based on the PCA loadings, PC1 captures age and number of products, which shows that the older customers are using fewer products. PC2 primarily represents credit score, which represents a customer's creditworthiness. PC3 is mainly influenced by gender.

After calculating principal components, we again check VIFs:

```
# Using PCA output to convert principal components to a DataFrame
pca_df = pd.DataFrame(principal_components, columns=['PC1', 'PC2', 'PC3'])
# Drop original multicollinear columns from the main dataset
df_clean = df1.drop(columns=multcol_cols)
# Merge PCA components with the cleaned dataset
final_df = pd.concat([df_clean, pca_df], axis=1)

# Again calculating the VIF's after PCA
vif_data = pd.DataFrame()
vif_data['Feature'] = final_df.columns
vif_data['VIF'] = [variance_inflation_factor(final_df.values, i) for i in
    range(final_df.shape[1])]

print(vif_data)
```

```
Output:
           Feature       VIF
0        Geography  3.858086
1           Tenure  3.442307
2        HasCrCard  3.328513
3   IsActiveMember  1.866942
4  EstimatedSalary  4.286402
5              PC1  1.003796
6              PC2  1.000145
7              PC3  1.000180
```

This result shows that, after applying PCA, the issue of multicollinearity has been significantly reduced. Initially, the variables Credit Score, Transformed Age, Gender, and NumOfProducts exhibited high Variance Inflation Factors (VIFs), indicating strong correlation among them. By transforming these variables into three principal components (PC1, PC2, and PC3), we successfully removed redundancy while preserving the most important information. In the next section, we will utilize Logistic Regression to analyze feature importance and understand which factors significantly influence customer churn.

6.3 Logistic Regression with Feature Importance

Logistic Regression is a statistical model commonly used for binary classification tasks, such as predicting customer churn. Unlike linear regression, which estimates continuous values, logistic regression estimates the probability of an outcome falling into one of two classes, typically using the sigmoid function:

$$P(Y=1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1 X_1+...+\beta_n X_n)}}$$

where β_0 is the intercept, $\beta_1, \beta_2, ..., \beta_n$ are coefficients for the independent variables $X_1, X_2, ...X_n$. The model optimizes these coefficients to minimize the classification error.

We trained a logistic regression model after handling multicollinearity using Principal Component Analysis (PCA):

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, classification_report

# Splitting for training and testing
X = final_df.drop("Churn", axis=1)
y = final_df["Churn"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
    random_state=42)


# Starndardizing
cols_to_stand = ["Geography", "Tenure", "HasCrCard",  "IsActiveMember", "
    EstimatedSalary"]
pc_columns = ["PC1", "PC2", "PC3"]
# Separate the columns to be standardized and the principal component
    columns
X_train_to_standardize = X_train[cols_to_stand]
X_train_pc = X_train[pc_columns]

scaler = StandardScaler()
X_standardized = scaler.fit_transform(X_train_to_standardize)
X_standardized = pd.DataFrame(X_standardized, columns=cols_to_stand, index
    =X_train.index) # Keep index

# Concatenate the standardized columns with the principal component
    columns
X_train_stand = pd.concat([X_standardized, X_train_pc], axis=1)


# Separate the columns to be standardized and the principal component
    columns
X_test_to_standardize = X_test[cols_to_stand]
X_test_pc = X_test[pc_columns]

scaler = StandardScaler()
X_test_standardized = scaler.fit_transform(X_test_to_standardize)
X_test_standardized = pd.DataFrame(X_test_standardized, columns=
    cols_to_stand, index=X_test.index) # Keep index
```

```

# Concatenate the standardized columns with the principal component
    columns
X_test_stand = pd.concat([X_test_standardized, X_test_pc], axis=1)


# Train logistic regression model
log_reg = LogisticRegression()
log_reg.fit(X_train_stand, y_train)


# Predict on test data
y_pred = log_reg.predict(X_test_stand)

# Evaluate accuracy and other metrics
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

This gives the evaluation:

```
Output:
Accuracy: 0.829841821377346
Classification  Report:
                precision   recall   f1-score   support

           0       0.85      0.96       0.90     39099
           1       0.69      0.35       0.46     10402

    accuracy                            0.83     49501
   macro avg       0.77      0.65       0.68     49501
weighted avg       0.81      0.83       0.81     49501
```

This implies that the model correctly predicts whether a customer will churn about 83% of the time. However, the recall[2] for class 1 is only 0.35, meaning the model identifies just 35% of actual churners. This is likely due to the imbalanced dataset, where there are significantly more non-churn customers than churn customers.

To improve recall for class 1, we balance the dataset using Synthetic Minority Oversampling Technique (SMOTE) as:

```
from imblearn.over_sampling import SMOTE
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, accuracy_score,
    confusion_matrix


smote = SMOTE(random_state=42)
X_train_smote, y_train_smote = smote.fit_resample(X_train_stand, y_train)
print(f"Before SMOTE: {y_train.value_counts()}")
print(f"After SMOTE: {pd.Series(y_train_smote).value_counts()}")
```

```
Output:
Before SMOTE: Churn
0    90988
1    24513
```

[2] Recall measures how many of the actual positive (churned) customers the model was able to correctly identify.

```
Name: count, dtype: int64
After SMOTE: Churn
0    90988
1    90988
Name: count, dtype: int64
```

```python
log_reg = LogisticRegression()
log_reg.fit(X_train_smote, y_train_smote)


# Predict on test data
y_pred = log_reg.predict(X_test_stand)

# Evaluate performance
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
Output:
Accuracy: 0.7303286802286822
Classification Report:
               precision    recall  f1-score   support

           0       0.91      0.73      0.81     39099
           1       0.42      0.74      0.54     10402

    accuracy                           0.73     49501
   macro avg       0.67      0.73      0.67     49501
weighted avg       0.81      0.73      0.75     49501
```

The accuracy has decreased from 83% to 73%, which is expected when balancing the dataset, as the model is now focusing more on detecting churners instead of just optimizing for the majority class. SMOTE improved recall from 35% to 74%, meaning the model is identifying a larger portion of actual churners. The overall F1-score for churners increased from 0.46 to 0.54, showing better balance between precision[3] and recall.

We find the coefficients and intercept as:

```python
coefficients = log_reg.coef_[0]
intercept = log_reg.intercept_
print("The coefficients are:", coefficients)
print("The intercept is:", intercept)
```

```
The coefficients are: [ 0.12980331 -0.05003956 -0.07797838 -0.569709
   0.04649561  0.98458983 0.04341899  0.0050123 ]
The intercept is: [-0.39781649]
```

Let,

$$\begin{aligned} Churn = &-0.3978 + 0.1298 \cdot Geography - 0.05004 \cdot Tenure \\ &- 0.078 \cdot HasCrCard - 0.5697 \cdot IsActiveMember \\ &+ 0.0465 \cdot EstimatedSalary + 0.9846 \cdot PC1 \\ &+ 0.0434 \cdot PC2 + 0.005 \cdot PC3 \end{aligned} \tag{6.1}$$

Therefore, the estimated sigmoid function is:

$$P(Y = 1|X) = \frac{1}{1 + e^{-Churn}} \tag{6.2}$$

[3]Precision tells us how many of the customers the model predicted as churners were actually churners.

After applying logistic regression to predict customer churn, we analyzed the importance of each feature using the model's coefficients. In logistic regression, the coefficients represent the impact of each feature on the likelihood of a customer churning, measured in log-odds[5]. A positive coefficient indicates that an increase in that feature increases the probability of churn, while a negative coefficient suggests a reduced likelihood of churn.

From our model, we observed that PC1 has the highest positive coefficient (0.98), indicating that it strongly increases the probability of churn. Since PC1 is derived from features NumOfProducts and Age, it likely represents similar pattern influencing customer behavior. Geography (0.13) and EstimatedSalary (0.04) also have positive coefficients, suggesting that customers from Spain region and those with higher estimated salaries may have a slightly higher tendency to churn.

On the other hand, IsActiveMember (-0.57) has the most significant negative impact, meaning that customers who are active members are much less likely to churn. This aligns with expectations, as active engagement with the bank often translates to better retention. Similarly, Tenure (-0.05) and HasCrCard (-0.07) have small negative coefficients, implying that longer relationships with the bank and having a credit card slightly reduce the chances of churn.

By applying principal components (PC1, PC2, and PC3), we effectively addressed multicollinearity while retaining the most important information. The results suggests that customer engagement, tenure, and certain demographic factors significantly influence churn behavior.

While logistic regression provided valuable insights into the key factors influencing customer churn, it has some limitations. Since it assumes a linear relationship between features and the target variable, it may not fully capture complex interactions. Additionally, logistic regression coefficients can be challenging to interpret when dealing with highly correlated features. To gain a deeper understanding of feature importance and explore non-linear relationships, we now apply tree-based models.

6.4 Feature Importance using Tree Based Models:

Understanding which factors influence customer churn is essential for making informed business decisions. While logistic regression provides valuable insights into feature importance through coefficients, tree-based models like XGBoost (Extreme Gradient Boosting) offer a more advanced way to evaluate feature significance. Unlike traditional models, XGBoost captures complex relationships between variables and assigns importance based on how often a feature is used in decision splits. This approach helps identify the most influential factors affecting customer churn, leading to better model interpretability and improved decision-making. In this section, we will use XGBoost to determine feature importance and analyze its impact on churn prediction.

```
# require libraies
import xgboost as xgb
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, accuracy_score

# Define features and target
X = final_df.drop("Churn", axis=1)
y = final_df['Churn']  # Target variable

# Split the data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
    random_state=42)
```

```python

# Create and train XGBoost classifier
xgb_model = xgb.XGBClassifier(
    learning_rate=0.1,
    n_estimators=100,
    max_depth=5,
    min_child_weight=1,
    gamma=0,
    subsample=0.8,
    colsample_bytree=0.8,
    objective='binary:logistic',
    nthread=4,
    scale_pos_weight=1,
    seed=42
)

# Handle class imbalance if needed
scale_pos_weight = len(y_train[y_train==0]) / len(y_train[y_train==1])
xgb_model.scale_pos_weight = scale_pos_weight

# Train the model
xgb_model.fit(X_train, y_train)

# Make predictions
y_pred = xgb_model.predict(X_test)

# Print metrics
print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print(f"Classification Report:\n{classification_report(y_test, y_pred)}")
```

```
Output:
Accuracy: 0.7948
Classification Report:
                precision   recall   f1-score   support

           0       0.93      0.80      0.86     39099
           1       0.51      0.79      0.62     10402

    accuracy                           0.79     49501
   macro avg       0.72      0.79      0.74     49501
weighted avg       0.84      0.79      0.81     49501
```

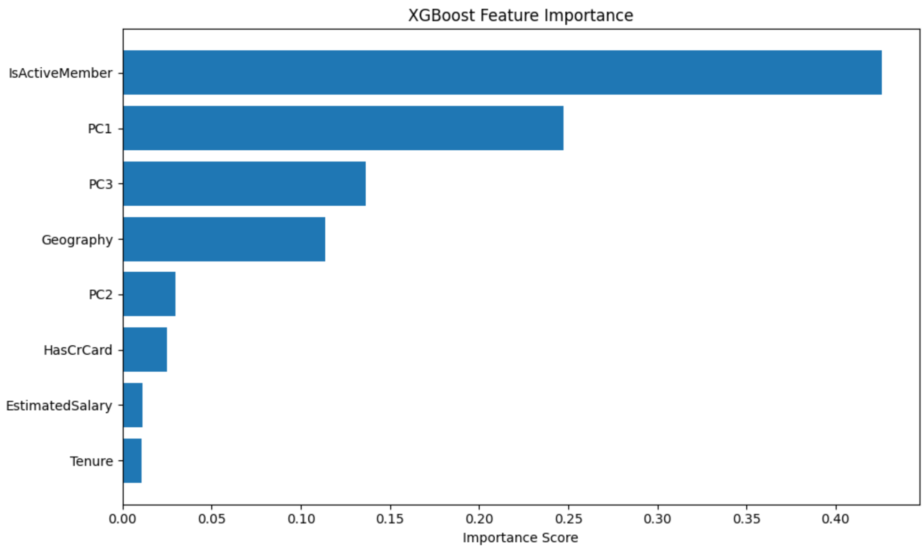
The XGBoost model achieved an accuracy of 79.48%, showing a reasonable improvement over logistic regression. More importantly, the recall for churners (customers labeled as '1') improved to 79%, meaning that the model correctly identified 79% of customers who are at risk of leaving the bank. This improvement in recall makes the model more suitable for addressing customer retention strategies.

After implementing the XGBoost classifier, we analyzed the importance of different features in predicting customer churn. XGBoost is a powerful tree-based model that assigns importance scores to features based on how frequently they are used in decision trees. This helps in understanding which features play a crucial role in identifying customers who are likely to leave the bank.

```python
# Get feature importance
importance = xgb_model.feature_importances_
feature_names = X.columns

# Create DataFrame for better visualization
```

```python
feature_importance = pd.DataFrame({
    ‘Feature’: feature_names,
    ‘Importance’: importance
}).sort_values(’Importance’, ascending=False)

print(feature_importance)

# Plot feature importance
plt.figure(figsize=(10, 6))
plt.barh(feature_importance[‘Feature’], feature_importance[‘Importance’])
plt.xlabel(‘Importance Score’)
plt.title(‘XGBoost Feature Importance’)
plt.gca().invert_yaxis()  # Display most important at the top
plt.tight_layout()
plt.show()
```

```
Output:
           Feature    Importance
3   IsActiveMember    0.425846
5              PC1    0.247357
7              PC3    0.136500
0        Geography    0.113615
6              PC2    0.029649
2        HasCrCard    0.024880
4  EstimatedSalary    0.011382
1           Tenure    0.010772
```



The feature importance scores assigned by XGBoost are as follows:

IsActiveMember is the most influential factor in predicting churn, with a high importance score of 0.4258. The first principal component (PC1) follows with 0.2473, indicating underlying patterns affecting customer churn. PC3 and Geography also contribute significantly, while features like PC2, HasCrCard, EstimatedSalary, and Tenure have lower importance scores.

Comparison with Logistic Regression: When comparing feature importance of Logistic Regression and XGBoost, the key differences are:

1. IsActiveMember is highly important in both models but has a stronger influence in XGBoost.

Feature	Importance Score
IsActiveMember	0.4258
PC1	0.2473
PC3	0.1365
Geography	0.1136
PC2	0.0296
HasCrCard	0.0248
EstimatedSalary	0.0113
Tenure	0.0108

2. PC1 and PC3 gain higher importance in XGBoost, capturing complex patterns missed by Logistic Regression.

3. Geography holds moderate importance in both models, showing its consistent effect on churn.

4. HasCrCard, EstimatedSalary, and Tenure remain less significant in both models.

While Logistic Regression provides direct interpretability through coefficients, XGBoost captures complex relationships more effectively, improving recall and churn prediction. The insights from XGBoost suggest that customer engagement (IsActiveMember) and behavioral trends (PC1, PC3) play the most critical roles in churn, which can help in implementing better retention stratergies.

6.5 Clustering Analysis using K-Means:

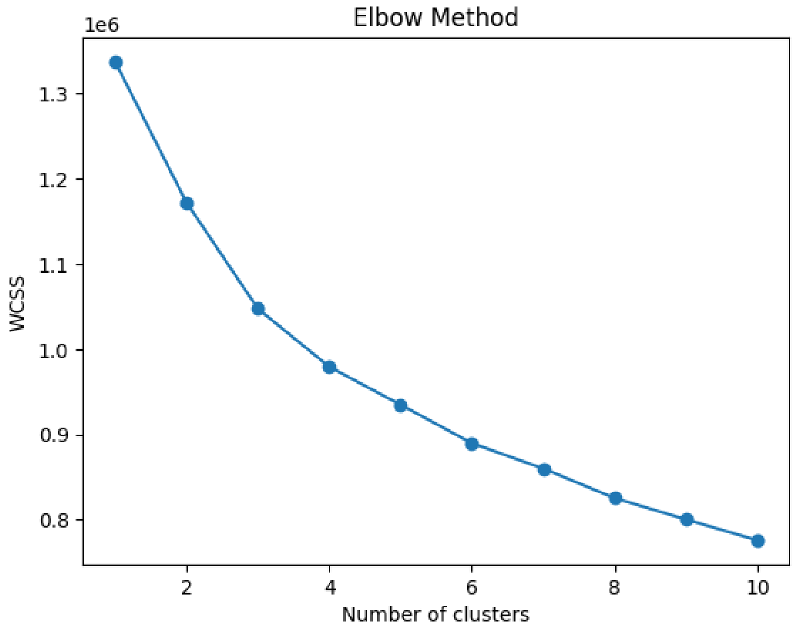
To better understand customer segments, we applied K-Means clustering, a widely used unsupervised learning technique. This method helps group customers with similar characteristics, providing insights for targeted marketing and customer retention strategies.

```python
# Standardizing the data
from sklearn.preprocessing import StandardScaler

# Splitting for training and testing
X = final_df.drop(columns=["Churn"], axis=1)
y = final_df["Churn"]

# Starndardizing
cols_to_stand = ["Geography", "Tenure", "HasCrCard",  "IsActiveMember", "
    EstimatedSalary"]
pc_columns = ["PC1", "PC2", "PC3"]
# Separate the columns to be standardized and the principal component
    columns
X_to_standardize = X[cols_to_stand]
X_pc = X[pc_columns]

scaler = StandardScaler()
X_standardized = scaler.fit_transform(X_to_standardize)
X_standardized = pd.DataFrame(X_standardized, columns=cols_to_stand, index
    =X.index) # Keep index

# Concatenate the standardized columns with the principal component
    columns
X_stand = pd.concat([X_standardized, X_pc], axis=1)
```

To determine the optimal number of clusters we used the Elbow Method, which plots the Within-Cluster Sum of Squares (WCSS) against different values of K (number of clusters).

```
# Determining optimal cluster using Elbow method
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Calculate WCSS (Within-Cluster Sum of Squares) for different k values
wcss = []
range_values = range(1, 11) # Looking for optimal k from 1 to 10
for i in range_values:
    kmeans = KMeans(n_clusters=i, init='k-means++', max_iter=300, n_init
    =10, random_state=0)
    kmeans.fit(X_stand)
    wcss.append(kmeans.inertia_)

# Plot the Elbow Method graph
plt.plot(range_values, wcss, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.show()
```

The output curve is:



The optimal cluster count was determined to be 3, as the curve showed an "elbow" at point 3.

```
# Applying K-means clustering
# Choose the optimal number of clusters
n_clusters = 3 # number of cluster determined using Elbow method

# Apply K-Means
kmeans = KMeans(n_clusters=n_clusters, init='k-means++', max_iter=300,
    n_init=10, random_state=0)
y_kmeans = kmeans.fit_predict(X_stand)

# Add cluster assignments to the original dataframe
final_df['Cluster'] = y_kmeans

# Analyze clusters
```

```
# Analyze cluster characteristics
cluster_summary = final_df.groupby‘Cluster’).mean()
print(cluster_summary)
```

```
Output:
        Geography       Tenure   HasCrCard   IsActiveMember
   EstimatedSalary   \
Cluster
0          1.672471   1822.869387        0.0          0.51617
   112182.911068
1          1.636838   1829.982345        1.0          1.00000
   112259.712881
2          1.644615   1841.093943        1.0          0.00000
   113133.645696

             PC1        PC2        PC3      Churn
Cluster
0         0.021417   0.007789  -0.001612   0.227408
1        -0.059322   0.005559   0.010243   0.116689
2         0.043638  -0.010379  -0.008875   0.293279
```

This gives the output as:

Cluster	Geography	Tenure	HasCrCard	IsActiveMember	EstimatedSalary	PC1	PC2	PC3	Churn Rate
0	1.67	1822.87	0.0	0.516	112182.91	0.021	0.007	-0.001	22.74%
1	1.63	1829.98	1.0	1.000	112259.71	-0.059	0.005	0.010	11.67%
2	1.64	1841.09	1.0	0.000	113133.64	0.043	-0.010	-0.008	29.33%

After applying K-Means clustering, we grouped customers into three different clusters based on their characteristics. Each cluster represents a group of customers who share similar behaviors. The intepretation of these cluster is as follows:

1. **Cluster 0: Moderately Engaged Customers (Churn Rate: 22.74%)**
    - Customers in this group are somewhat active, but not fully engaged. Their IsActiveMember score is 0.516, meaning about half of them are active members while the other half are not.
    - They do not hold a credit card (HasCrCard = 0), which might indicate that they are either new customers or do not use additional banking services.
    - Their tenure (time with the bank) is moderate, and their estimated salary is similar to the other clusters.
    - Since around 22.74% of customers in this group are leaving, they are at a moderate risk of churn. The bank could try to increase their engagement by offering personalized banking services or introducing incentives to make them more loyal.
2. **Cluster 1: Highly Loyal Customers (Churn Rate: 11.67%)**
    - This is the most stable group with the lowest churn rate (11.67%).
    - These customers are highly active members (IsActiveMember = 1.0), meaning they regularly interact with the bank's services.
    - They also own a credit card (HasCrCard = 1.0), which suggests they are more financially involved with the bank.
    - Their tenure is slightly higher than Cluster 0, and their salary is in the same range.

- Since these customers are already engaged and loyal, the bank should focus on retaining them through loyalty programs, premium benefits, and exclusive offers.

3. **Cluster 2: High-Risk Customers (Churn Rate: 29.33%)**
    - This group has the highest churn rate (29.33%), meaning nearly one in three customers in this group is leaving.
    - A key reason for this could be that they are not active members (IsActiveMember = 0), meaning they rarely use the bank's services.
    - Unlike Cluster 0, they do have credit cards, but that does not seem to be enough to keep them engaged.
    - Their tenure and salary are similar to the other clusters, so these factors do not play a big role in their decision to leave.
    - To reduce churn in this group, the bank should focus on re-engaging these customers. They could send personalized offers, discounts, or reminders about unused services to bring them back into active banking.

The Clustering Analysis using K-Means helped us identify distinct customer segments based on important features. By analyzing these clusters, we observed differences in customer activity, financial behavior, and churn risk, which can guide targeted retention strategies. With this, we complete the Multivariate Analysis.

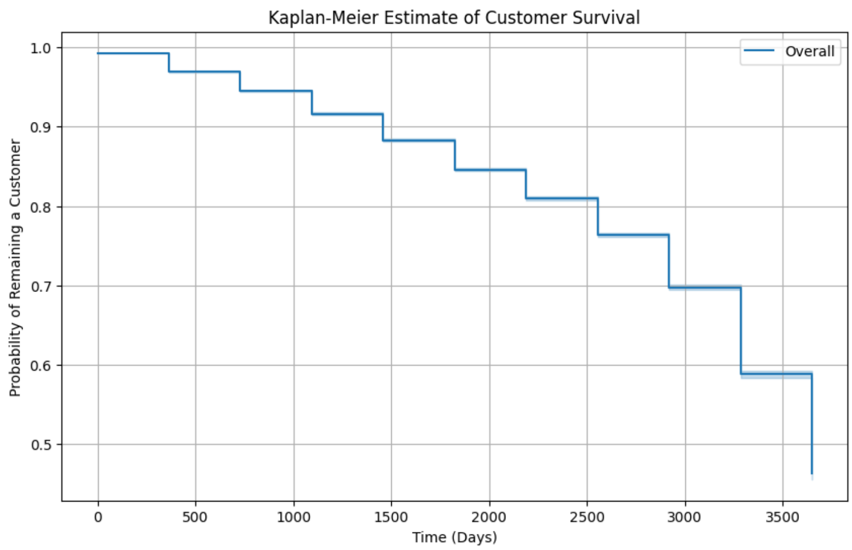
Chapter 7
Survival Analysis

Survival analysis is a statistical method used to study the time until an event of interest happens in this case, customer churn. Unlike regular classification methods, it not only helps predict whether a customer will leave but also estimates when they might churn. This time-based approach gives deeper insights into customer behavior and helps businesses plan timely interventions. In this chapter, we apply survival analysis techniques to understand the risk and timing of customer churn more effectively.

7.1 Kaplan-Meier Estimation:

The Kaplan-Meier estimator is a non-parametric method used to estimate the survival function from observed data. It helps us understand the probability that a customer remains active at different time intervals. The following KM survival curve provides an overall view of customer survival over time.

```
from lifelines import KaplanMeierFitter
T = df['Tenure']  # Time
E = df['Exited']  # Event indicator (1 = churned, 0 = still active)
# Fit the KM model
kmf = KaplanMeierFitter()
kmf.fit(T, event_observed=E, label='Overall')
# Plot the survival curve
plt.figure(figsize=(10, 6))
kmf.plot_survival_function()
plt.title('Kaplan-Meier Estimate of Customer Survival')
plt.xlabel('Time (Days)')
plt.ylabel('Probability of Remaining a Customer')
plt.grid(True)
plt.show()
```

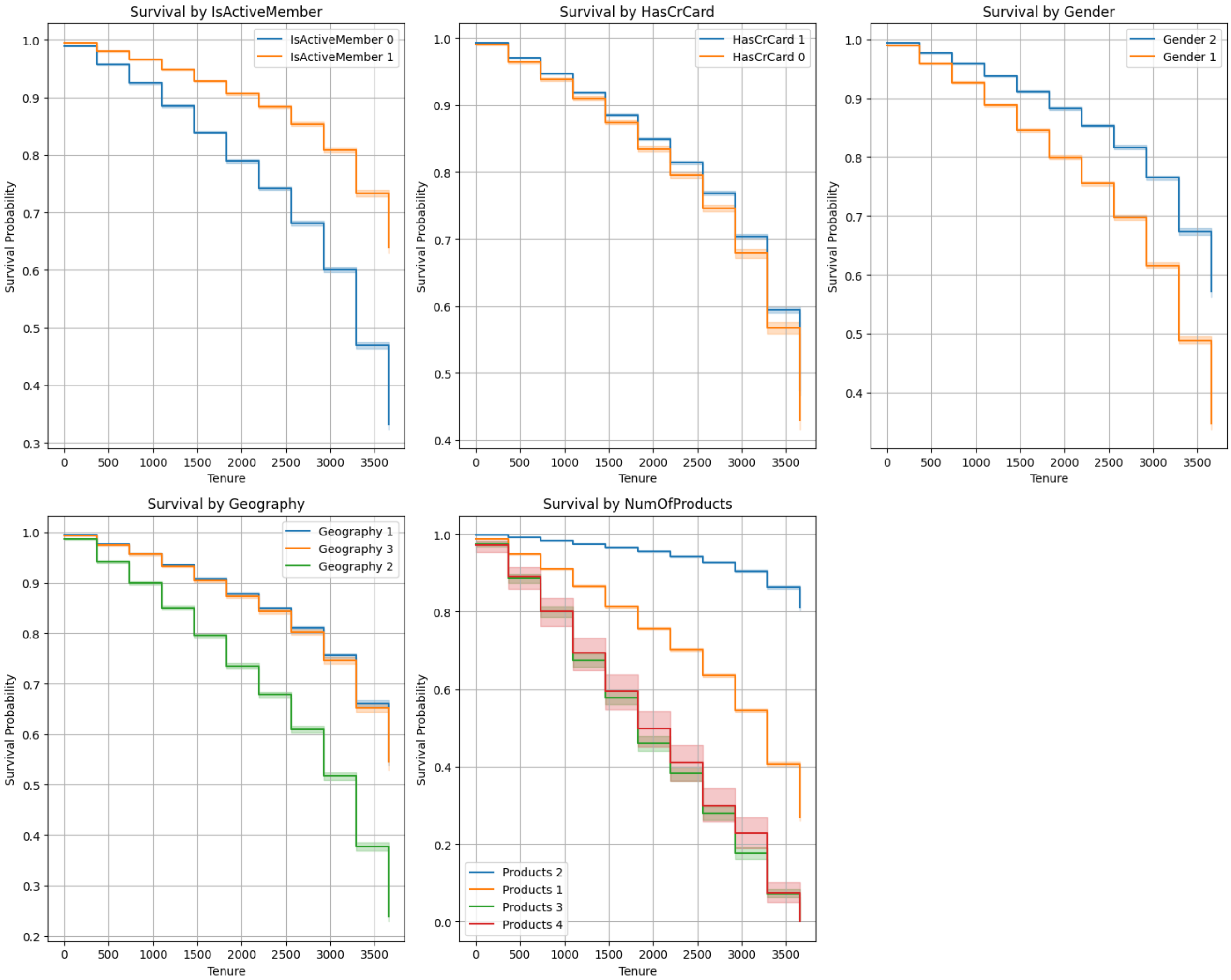


The survival curve shows a gradual decline in customer retention over time, indicating that as tenure increases, the likelihood of customers remaining with the bank decreases.

After analyzing the overall survival curve using the Kaplan-Meier estimator, we further explore how different customer characteristics influence survival probability. This helps us understand which factors contribute to customer churn.

We analyze survival rates based on five key categorical features:

```
categorical_cols = [‘IsActiveMember’, ‘HasCrCard’, ‘Gender’, ‘Geography’, ‘NumOfProducts’]

plt.figure(figsize=(15, 12))

for i, col in enumerate(categorical_cols):
    plt.subplot(2, 3, i + 1)

    for category in df[col].unique():
        subset = df[df[col] == category]
        T = subset[‘Tenure’]
        E = subset[‘Exited’]

        kmf = KaplanMeierFitter()
        if col == ‘Gender’:
          label = f‘Gender {category}’
        elif col == ‘Geography’:
          label = f‘Geography {category}’
        elif col == ‘NumOfProducts’:
          label = f‘Products {category}’
        else:
          label = f‘{col} {category}’
        kmf.fit(T, event_observed=E, label=label)
        kmf.plot_survival_function(ax=plt.gca())

    plt.title(f‘Survival by {col}’)
    plt.xlabel(‘Tenure’)  # Adjust as needed
    plt.ylabel(‘Survival Probability’)
    plt.grid(True)
plt.tight_layout()
plt.show()
```



Interpretation:

1. **IsActiveMember:** Customers who are active members (marked as 1) have a higher survival probability compared to inactive members. This suggests that customers who engage more with the bank's services are less likely to leave. In contrast, inactive members (marked as 0) tend to churn at a much faster rate, meaning they are more likely to stop using the bank's services over time.
   Business Insight: Encouraging customers to stay engaged such as through loyalty programs or personalized offers might improve retention.

2. **HasCrCard:** The survival curves for customers with and without a credit card are quite similar, suggesting that credit card ownership does not have a strong impact on customer retention. This indicates that just having a credit card does not necessarily make a customer more loyal to the bank.
   Business Insight: Simply offering credit cards might not improve retention. Instead, providing additional benefits (e.g. cashback, rewards) might help.

3. **Gender:** The survival probability follows a similar downward trend for both male and female customers. While there might be slight differences in churn rates between genders, the overall impact is relatively small.
   Business Insight: Gender alone does not seem to be a strong predictor of churn, meaning retention strategies should focus on other behavioral factors rather than gender-based targeting.

4. **Geography:** Customers from different regions (Geography 1, Geography 2, and Geography 3) show different survival patterns. One region (likely Geography 2 in the plot which is Germany) has a noticeably lower survival probability, meaning customers from that region are more likely to churn faster. This could be due to regional economic factors, service availability, or local competition from other banks.
   Business Insight: Banks should consider region-specific marketing strategies or improve services in areas where churn is higher.

5. **NumOfProducts:** Customers who own 1 and 4 products (minimum and maximum number of products) tend to churn much faster than those with multiple products. The survival curve shows a steep decline for customers with just one product, while those with two or three products have a significantly higher retention rate. This suggests that customers who engage with two or three products (e.g., loans, savings accounts, credit cards) stay more connected to the bank and are less likely to leave.
   Business Insight: Banks can improve retention by cross-selling additional products to customers with only one product.

7.2 Log-Rank Test:

The Log-Rank test is a non-parametric statistical test used to compare the survival distributions of two or more groups. It assesses whether there is a significant difference in the probability of survival over time between different categories of a categorical variable.

In survival analysis, the Log-Rank test is particularly useful when comparing customer churn behavior across different groups, such as gender, geography, or other demographic factors. The test is based on Kaplan-Meier survival estimates and compares the observed vs. expected number of events (churn) at each time point.

For the categorical features we test whether the survival distributions of different groups are statistically different using the hypothesis:

H_0 : There is no significant difference in the survival distributions across the categories of the feature.

H_1 : There is a significant difference in the survival distributions across the categories of the feature.

If the p-value is below the significance level $\alpha = 0.05$, we reject the null hypothesis, concluding that the categorical feature significantly influences survival time (customer tenure).

Summary of Log-Rank test results is in table 7.1.

We can summarize the results as:

1. **Impact of Customer Activity (IsActiveMember):** The survival curves of active and inactive customers are significantly different (p-value < 0.005). This suggests that active customers tend to stay with the bank longer, while inactive members have a higher likelihood of churning.

2. **Impact of Credit Card Ownership (HasCrCard):** A significant difference in survival curves was observed between customers who own a credit card and those who do not (p-value < 0.005). Customers without a credit card have a higher churn rate, indicating that credit card ownership might contribute to customer retention.

3. **Impact of Gender:** The log-rank test detected a significant difference between the survival curves of male and female customers (p-value < 0.005). This indicates that gender does influence customer churn.

Comparison	Test Statistic	p-value	Significant?
IsActiveMember (Yes vs No)	5961.96	<0.005	**Yes**
HasCreditCard (Yes vs No)	74.59	<0.005	**Yes**
Gender (Male vs Female)	3084.38	<0.005	**Yes**
Geography (1 vs 2)	5548.56	<0.005	**Yes**
Geography (1 vs 3)	6.14	0.01	**Yes**
Geography (2 vs 3)	3038.24	<0.005	**Yes**
NumOfProducts (1 vs 2)	195.05	<0.005	**Yes**
NumOfProducts (1 vs 3)	2504.62	<0.005	**Yes**
NumOfProducts (1 vs 4)	382.41	<0.005	**Yes**
NumOfProducts (2 vs 3)	22556.28	<0.005	**Yes**
NumOfProducts (2 vs 4)	4731.92	<0.005	**Yes**
NumOfProducts (3 vs 4)	1.57	0.21	**No**

Table 7.1: Log-Rank Test Results for Categorical Variables

4. **Impact of Geography:** Pairwise comparisons between different geographical locations revealed significant differences. Customers in Geography 1 (France) vs Geography 2 (Germany) (p-value < 0.005) and Geography 2 (Germany) vs Geography 3 (Spain) (p-value < 0.005) show distinct survival patterns. However, the difference between Geography 1 (France) and Geography 3 (Spain) is weaker (p-value $= 0.01$), though still below the 0.05 threshold. This suggests that regional differences influence customer retention, possibly due to variations in banking services, economic conditions, or cultural factors.

5. **Impact of Number of Products Held:** Customers with different numbers of products exhibit significantly different survival patterns. Notably customers with 1 vs. 2 products and 2 vs. 3 products show highly significant differences (p-value < 0.005). However, the difference between customers with 3 vs. 4 products is not significant (p-value $= 0.21$), indicating that additional products beyond three may not strongly influence retention. This suggests that holding more banking products (up to 3) is associated with longer retention, but further increasing the number of products may not significantly impact churn.

The survival analysis reveals that customer activity, credit card ownership, gender, geography, and the number of products held are significant factors influencing churn at the 5% significance level. These insights can help the bank tailor its customer retention strategies by focusing on engaging inactive customers, promoting credit card usage, addressing geographical disparities, and encouraging customers to use multiple products.

Chapter 8
Conclusion

This project focused on analyzing and predicting customer churn in the banking sector using statistical and machine learning techniques. The main aim was to uncover the key factors that influence customer attrition and to develop insights that can help banks retain their customers more effectively. Throughout the study, we explored the dataset through various stages, starting from initial data exploration to building predictive models. The study led to the following findings:

1. While checking for normality, we found that the feature Age was right-skewed. To correct this, we applied the Box-Cox transformation with $\lambda = -0.24589$, which helped normalize the distribution for further analysis.

2. Through Exploratory Data Analysis (EDA), we observed that customers with shorter tenure, fewer products, and inactive account status were more likely to churn. We also found that geography and gender played a role in influencing customer churn.

3. Using Variance Inflation Factors (VIFs), we detected severe multicollinearity in features such as Age, CreditScore, and Gender. The feature NumOfProducts also showed notable multicollinearity. To address this issue, we applied Principal Component Analysis (PCA), which helped reduce redundancy in the data. Three principal components were extracted that explained the maximum variance and were used in further analysis.

4. We trained two predictive models: Logistic Regression (with class imbalance handled using SMOTE) and XGBoost. The results showed that XGBoost outperformed Logistic Regression in predicting churn. Comparing feature importance from both models revealed that IsActiveMember was the most influential feature in both, with stronger influence in XGBoost. PC1 and PC3 also showed higher importance in XGBoost, capturing complex patterns not identified by Logistic Regression. Geography maintained moderate importance in both models, while features like HasCrCard, EstimatedSalary, and Tenure were less significant.

5. Using K-means clustering, we segmented the customers into three distinct groups: moderately engaged customers, highly active customers, and high-risk customers. This clustering provided deeper insights into customer behavior and allowed for the identification of target groups for retention strategies.

6. Kaplan-Meier survival curves and Log-rank tests showed that all categorical features (IsActiveMember, HasCrCard, Gender, Geography, and NumOfProducts) significantly affected churn. These insights confirm that certain demographic and behavioral traits are closely linked with customer attrition.

Based on these findings, banks can take several important steps to improve customer retention. First, they should monitor high-risk customer segments more closely and provide them with personalized engagement offers. Customers who are inactive or have short tenure should be prioritized for loyalty programs or special benefits to increase their satisfaction and retention. Additionally, geographic regions and gender groups showing higher churn tendencies should receive targeted marketing and support. Banks can also build internal churn prediction tools using machine learning models like XGBoost, which can provide real-time alerts about potential churners. Finally, regularly assessing customer data for patterns and trends can help banks stay proactive in their customer retention efforts.

The colab notebooks for all the codes used in the study are:

1. For pre-processing: `https://drive.google.com/file/d/1PPDVkmMpwuBRUZWtJ_HrewOXK19YwmVW/view?usp=sharing`
2. For EDA : `https://colab.research.google.com/drive/1Vl56uk6lD8QxuOrshDScIpl-2xi79dBe?usp=sharing`
3. For multivariate analysis: `https://colab.research.google.com/drive/1dCax8v_WcAm3B2ZbzKsUPa9H9UL8H6LX?usp=sharing`
4. For Survival analysis: `https://colab.research.google.com/drive/1Ey9OsNPadwHYKTJvUdLE0U007a1AP2ST?usp=sharing`

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